I. Completions (5 items, 3 points each)

Score	47

1. $\lim\limits_{n\to\infty}\sqrt[n]{2018^n+2019^n+2020^n}=$ 2020.

2. The minimum positive period of the function $f(x)=\sin\frac{2}{3}x+\cos\frac{3}{2}x$ is 12π.

$T=3\pi$ $T=3$

3. If $(1-\cos x)\ln(1+x^2)=o(x\sin x^n)(x\to 0)$, $x\sin x^n=o(e^{x^2}-1)$ $(x\to 0)$, then positive integer $n=$ 2.

4. If $\lim\limits_{x\to+\infty}\left(\frac{x^2+1}{x+1}-ax-b\right)=0$, then $a=$ 1, $b=$ -1.

5. $\lim\limits_{x\to 0}\left(\frac{2+e^{\frac{1}{x}}}{1+e^{\frac{1}{x}}}+\frac{\sin x}{|x|}\right)=$ 1.

II. Choices (5 items, 3 points each)

Score	

6. The correct one in the following propositions is (B)

A. Each periodic function must have a minimum positive period.

B. $[x]$ is a periodic function.

C. $f(x)=\begin{cases}x & x\geq 0\\ -x & x<0\end{cases}$ is not an elementary function.

D. $x\cos x$ is not a periodic function.

7. Find the rational conclusion of the function $f(x)=x\cos x$ as $x\to\infty$ (C)

A. infinite. B. infinitesimal.

C. Unbounded function but not infinite. D. Bounded function but not infinitesimal.

8. Let $f(x)=\begin{cases}x\sin\frac{1}{x} & x\neq 0\\ 0 & x=0\end{cases}$, which one is correct for $f(x)$ at $x=0$ (B)

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A. limit exists. B. $f(x)$ is continuous at $x=0$.

C. limit exists but it is not continuous. D. unconfirmed.

9. Let $f(x)=\begin{cases}\arctan\frac{1}{x} & x\neq 0\\ 0 & x=0\end{cases}$, then $x=0$ is (A)

A. a jump point. B. a removable discontinuity point.

C. an infinite point. D. a point of continuity.

10. Let $f(x)$ be defined on R, then (B)

A. $f(x)$ is continuous if and only if $f^2(x)$ is continuous.

B. $f(x)$ is continuous if and only if $f^3(x)$ is continuous.

C. $f(x)$ is continuous if and only if $|f(x)|$ is continuous.

D. $f(x)$ is continuous if and only if $f(f(x))$ is continuous.

III. Calculations and Proof (7 items, 8 points each)

Score	

11. Proving $\lim\limits_{n\to\infty}\frac{n!}{n^n}=0$ using the concept of limit of the numerical sequence.

12. Let $\{a_n\}$ be a numerical sequence, $a_1=\sqrt{2}$, $a_{n+1}=\sqrt{2a_n}$,

(1) Determine if the limit of $\{a_n\}$ exists:

(2) If the limit of $\{a_n\}$ exists, please give your answer.

13. Calculating $\lim\limits_{x\to+\infty}\left(x+\sqrt{x^2+100}\right)$.

14. Calculating $\lim\limits_{x\to 0}\frac{\ln(\cos x)}{x(e^{2x}-1)}$.

15. Calculating $\lim\limits_{x\to 0}\left(\frac{1+\tan x}{1+\sin x}\right)^{\frac{1}{x^2}}$.

16. Calculating $\lim\limits_{x\to 0}x\left[\frac{1}{x}\right]$.

17. Let $f(x)=\lim\limits_{n\to\infty}\frac{1-2^{nx}}{1+2^{nx}}$, find the discontinuous point of the function and determine the type.

IV. Analytical calculations (7 points)

Score	

18. Let $\lim\limits_{x\to 0}\frac{\sqrt{1+f(x)\sin x^2}-1}{1-\cos x}=3$, find $\lim\limits_{x\to 0}f(x)$.

I. Completions (5 items, 3 points each)

1. 2020 **2.** 12π **3.** 2 **4.** 1 -1 **5.** 1

II. Choices (5 items, 3 points each)

1. D **2.** C **3.** B **4.** A **5.** B

11. Proof: $\because \dfrac{n!}{n^n}=\dfrac{n\cdot(n-1)\cdots 1}{n\cdot n\cdots n}<\dfrac{1}{n}$,

For any $\varepsilon>0$, to ensure $|a_n-a|=\left|\dfrac{n!}{n^n}-0\right|=\dfrac{n!}{n^n}<\varepsilon$,

We just need $\dfrac{1}{n}<\varepsilon$, i.e., $n>\dfrac{1}{\varepsilon}$, we can take $N=\left[\dfrac{1}{\varepsilon}\right]$,

$|a_n-a|=\left|\dfrac{n!}{n^n}-0\right|<\varepsilon$ holds, if $n>N$,

$\therefore \lim\limits_{n\to\infty}\dfrac{n!}{n^n}=0$.

12. (1) We prove that $\{a_n\}$ is bounded firstly.

$\because 0<a_1<2$, Suppose $0<a_k<2$, then $0<a_{k+1}=\sqrt{2a_k}<\sqrt{2\cdot 2}=2$,

$\therefore 0<a_n<2 \quad n=1,2,\cdots$.

The monotonicity of $\{a_n\}$ is proved as follows.

$\because a_{n+1}-a_n=\sqrt{2a_n}-a_n=\dfrac{(2-a_n)a_n}{\sqrt{2a_n}+a_n}>0$, $\therefore \{a_n\}$ is an increasing sequence.

So the limit of $\{a_n\}$ exists. 5 points

(2) Assume $\lim\limits_{n\to\infty}a_n=a$, then $\lim\limits_{n\to\infty}a_n=\lim\limits_{n\to\infty}\sqrt{2a_n}$, i.e., $a=\sqrt{2a}$

$\Rightarrow a=0\,(dropping)$ or $a=2 \Rightarrow a=2$. 8 points

13. Set $t=-x\to+\infty$, then

$$\lim_{x\to-\infty}\left(x+\sqrt{x^2+100}\right)=\lim_{t\to+\infty}\left(\sqrt{t^2+100}-t\right)=\lim_{t\to+\infty}\frac{100}{\sqrt{t^2+100}+t}=0$$

14. $$\lim_{x\to 0}\frac{\ln(\cos x)}{x(e^{2x}-1)}=\lim_{x\to 0}\frac{\ln\left[1+(\cos x-1)\right]}{x(e^{2x}-1)}=\lim_{x\to 0}\frac{\cos x-1}{x\cdot 2x}$$

$$=\lim_{x\to 0}\frac{-x^2/2}{2x^2}=-\frac{1}{4}$$

15. $$\lim_{x\to 0}\left(\frac{1+\tan x}{1+\sin x}\right)^{\frac{1}{x^2}}=\lim_{x\to 0}\left(1+\frac{\tan x-\sin x}{1+\sin x}\right)^{\frac{1}{x^2}}$$

$$=\lim_{x\to 0}\left(\left(1+\frac{\tan x-\sin x}{1+\sin x}\right)^{\frac{1+\sin x}{\tan x-\sin x}}\right)^{\frac{\tan x-\sin x}{1+\sin x}\cdot\frac{1}{x^2}}=e^{\lim\limits_{x\to 0}\frac{\tan x(1-\cos x)}{x^2}}=e^{\frac{1}{2}}$$

16. $\because \dfrac{1}{x}-1<\left[\dfrac{1}{x}\right]\le\dfrac{1}{x}$ *if* $x\ne 0$

$\therefore 1-x<x\left[\dfrac{1}{x}\right]\le 1$ *if* $x>0$; $1\le x\left[\dfrac{1}{x}\right]<1-x$ *if* $x<0$;

Since $\lim\limits_{x\to 0^+}(1-x)=1$ *and* $\lim\limits_{x\to 0^-}(1-x)=1$,

The Sandwich Theorem implies that $\lim\limits_{x\to 0^+}x\left[\dfrac{1}{x}\right]=1$ *and* $\lim\limits_{x\to 0^-}x\left[\dfrac{1}{x}\right]=1$

Thus $\lim\limits_{x\to 0}x\left[\dfrac{1}{x}\right]=1$.

17. $\because f(x)=\lim\limits_{x\to 0}\dfrac{1-(2^x)^n}{1+(2^x)^n}=\begin{cases}1 & x<0\\ 0 & x=0\\ -1 & x>0\end{cases}$

$\therefore \lim\limits_{x\to 0^+}f(x)=-1\ne\lim\limits_{x\to 0^-}f(x)=1$, So $x=0$ is a jump point.

18. $\because \lim\limits_{x\to 0}\dfrac{\sqrt{1+f(x)\sin x^2}-1}{1-\cos x}=3$

$\therefore \sqrt{1+f(x)\sin x^2}-1\sim 3(1-\cos x)\sim\dfrac{3}{2}x^2\ (x\to 0)$

$\because \sqrt{1+f(x)\sin x^2}-1\sim\dfrac{1}{2}f(x)\sin x^2\sim\dfrac{1}{2}f(x)x^2$,

$\therefore \dfrac{1}{2}f(x)x^2\sim\dfrac{3}{2}x^2\ (x\to 0)$

$\therefore \lim\limits_{x\to 0}f(x)=3$.